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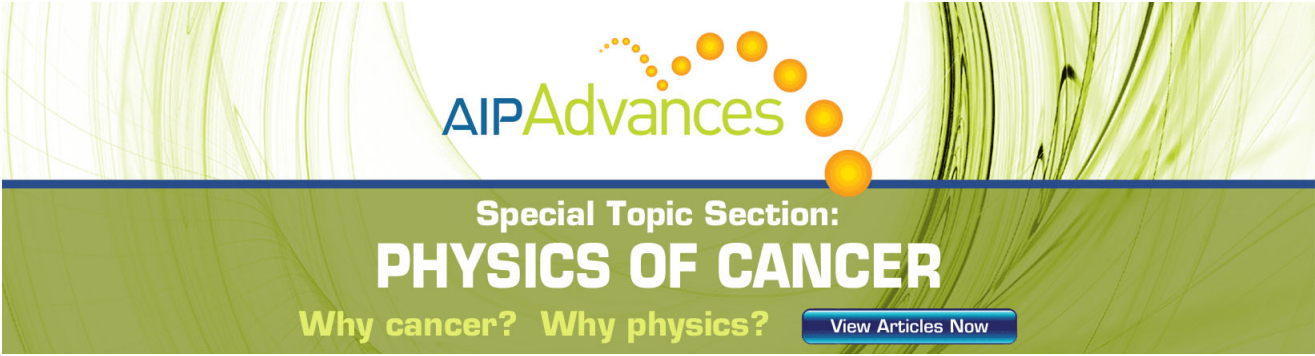
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# Polyatomic canonical variational theory for chemical reaction rates. Separable-mode formalism with application to $\text{OH} + \text{H}_2 \rightarrow \text{H}_2\text{O} + \text{H}$

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A formalism for the application of variational transition-state theory and semiclassical vibrationally adiabatic transmission coefficients to bimolecular reactions involving an arbitrary number of atoms is presented. This generalizes previous work on atom-diatom reactions. We make applications in this paper to the reactions  $\text{OH} + \text{H}_2 \rightarrow \text{H}_2\text{O} + \text{H}$  and  $\text{OH} + \text{D}_2 \rightarrow \text{HDO} + \text{D}$  using the Schatz-Elgersma fit to the Walch-Dunning *ab initio* potential energy surface. For both reactions we find large differences between conventional and variational transition-state theory and large effects of anharmonicity on the calculated rate constants. The effect of reaction-path curvature on the calculated transmission coefficients and rate constants is also large. The final calculated values of the kinetic isotope effects are in good agreement with experiment at high temperature but too large at room temperature.

## I. INTRODUCTION

Canonical variational transition-state theory (which we abbreviate as canonical variational theory or CVT) has proved to be a practical method for calculating thermal rate constants for atom-diatom reactions.<sup>1-7</sup> Conventional transition-state theory may be interpreted as a calculation of the free energy of activation under the assumption that a configuration-space dividing surface located at the highest-potential-energy point on the minimum-potential-energy path (MEP) provides a perfect bottleneck to reaction, i.e., that recrossing effects are negligible for such a dividing surface. In CVT we calculate the free energy of activation under the assumption that a configuration-space dividing surface located at the highest-free-energy-of-activation point on the MEP provides a perfect bottleneck. Thus, CVT is more consistent than conventional transition-state theory, in that it incorporates entropic and zero-point effects in locating the dividing surface as well as in calculating the rate constant for a given dividing surface. Furthermore, atom-diatom studies have shown that CVT is more accurate.<sup>6,7</sup> In the present article we extend the CVT to the treatment of bimolecular reactions of polyatomic species. An important ingredient in the general formulation presented here is the reaction-path Hamiltonian of Miller, Handy, and Adams.<sup>8</sup>

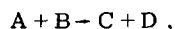
We illustrate the new methods by application to the reaction



using the analytic potential energy surface obtained by Schatz and Elgersma<sup>9</sup> by a fit to the *ab initio* calculations of Walch and Dunning.<sup>10</sup> The calculated rate constants and kinetic isotope effects are compared to the recent experimental results of Ravishankara and co-workers.<sup>11</sup>

## II. THEORY

Consider the general reaction



where any of A, B, C, and D may be atoms or diatoms or polyatomic molecules, with masses  $m_A$ ,  $m_B$ , etc.

Assume there are a total of  $N$  atoms (with  $N_A$  in A, etc.) so that

$$N = N_A + N_B = N_C + N_D. \quad (1)$$

Consider a scaled coordinate system defined by

$$x_{i\gamma} = (m_i/\mu)^{1/2}(R_{i\gamma} - R_{c.m.,\gamma}), \quad (2)$$

$i = 1, 2, \dots, N; \gamma = x, y, z,$

where  $R_{i\gamma}$  and  $R_{c.m.,\gamma}$  are laboratory-frame Cartesian coordinates of atom  $i$  and the system center of mass, respectively,  $m_i$  is the mass of atom  $i$ , and  $\mu$  is the reduced mass of reactant relative translational motion

$$\mu = \frac{m_A m_B}{m_A + m_B}. \quad (3)$$

The motivation for this choice of coordinate system is that the kinetic energy simplifies from

$$T = \frac{1}{2} \sum_{i=1}^N m_i \sum_{\gamma} \dot{R}_{i\gamma}^2 \quad (4)$$

to

$$T = \frac{1}{2} \mu \sum_{i=1}^N \sum_{\gamma} \dot{x}_{i\gamma}^2, \quad (5)$$

where an overdot denotes a time derivative. The fact that the reduced mass is the same for all coordinates means that a path of steepest descents in the  $\mathbf{x}$  coordinate system corresponds to an intrinsic reaction coordinate (IRC), i.e., to the solution of Hamilton's equations for a system whose instantaneous motion is continuously damped to an infinitesimal speed.<sup>12</sup> This removes some of the arbitrariness from the minimum energy path (MEP)<sup>13</sup> and the generalized transition states<sup>14</sup> to be defined next. The minimum energy path (MEP) is defined by joining the IRC from the saddle point to the reactants to the IRC from the saddle point to the products. (For a system with more than one saddle point, one must trace IRC's in both directions from each saddle point until one reaches either a local minimum or an asymptotic region; one can then join these IRC's to make the whole MEP.<sup>15</sup>) A generalized transition state is any phase-space hypersurface that divides reactants from

products.<sup>16,17</sup> In the present implementation of generalized transition-state theory (GTST), we consider only coordinate-space dividing surfaces that, near their intersection with the MEP, are hyperplanes in the  $\mathbf{x}$  coordinate system, and we require them to be perpendicular to the MEP at the point of intersection. Far from the intersection with the MEP, we require the hypersurfaces to bend if necessary so that they truly divide reactants from products at all total energies up to some high energy  $D$  (the specific choice we make for  $D$  is discussed in Sec. III A), and we consider only temperatures where the contributions to the reaction rate and to all partition functions from energies higher than  $D$  are negligible. We further define  $s$  as the arc length along the MEP from the highest saddle point (or the earliest if there are two or more of identical height), so that  $s$  values of  $-\infty$ ,  $0$ , and  $\infty$  refer to reactants, saddle point, and products, respectively.

Then the GTST dividing surfaces are parametrized by the value of  $s$  at which they intersect the MEP. For any GTST dividing surface, the GTST rate constant is the equilibrium one-way flux coefficient of phase-space trajectories through the dividing surface in the product direction, i.e., it is the net density of forward-crossing states per unit time divided by the density of reactant states per unit volume. In the same way as derived in detail elsewhere for collinear atom-diatom reactions,<sup>14,17</sup> the GTST rate constant for temperature  $T$  and dividing surface at  $s$  may be reduced to

$$k^{\text{GT}}(T, s) = \frac{\sigma}{\beta h} \frac{Q^{\text{GT}}(T, s)}{\Phi^{\text{R}}(T)} \exp[-\beta V_{\text{MEP}}(s)], \quad (6a)$$

$$= \frac{\sigma}{\beta h} K^0 \exp[-\beta \Delta G^{\text{GT},0}(T, s)], \quad (6b)$$

where we have introduced the following notation:  $\sigma$  is the symmetry factor<sup>7,18</sup> (number of equivalent paths);  $\beta$  is  $(k_{\text{B}}T)^{-1}$ , where  $k_{\text{B}}$  is Boltzmann's constant;  $h$  is Planck's constant;  $V_{\text{MEP}}(s)$  is the classical potential energy on the MEP at  $s$  with overall zero of energy at the reactants' classical equilibrium position, i.e.,  $V_{\text{MEP}}(s = -\infty) = 0$ ;  $Q^{\text{GT}}(T, s)$  is the partition function for the dividing surface at  $s$  with zero of energy for this partition function at  $V_{\text{MEP}}(s)$  and with all rotational symmetry numbers set to unity;  $\Phi^{\text{R}}(T)$  is the reactants' partition function per unit volume with the zero of energy at the overall zero of energy and with all rotational symmetry numbers set equal to unity;  $K^0$  is the reciprocal of the standard-state concentration (taken here as  $1 \text{ cm}^3 \text{ molecule}^{-1}$ ); and  $\Delta G^{\text{GT},0}(T, s)$  is the generalized standard-state free energy of activation for the dividing surface at  $s$ .

The CVT rate constant  $k^{\text{CVT}}(T)$  is obtained by locating the dividing surface variationally to maximize  $\Delta G^{\text{GT},0}(T, s)$ . The resulting dividing surface location is called  $s_{\text{CVT}}^{\text{GT}}(T)$ ; hence,

$$k^{\text{CVT}}(T) = \min_s k^{\text{GT}}(T, s), \quad (7a)$$

$$= k^{\text{GT}}[T, s_{\text{CVT}}^{\text{GT}}(T)]. \quad (7b)$$

In the above formulation,  $Q^{\text{GT},0}(T, s)$ ,  $\Phi^{\text{R}}(T)$ , and  $\Delta G^{\text{GT},0}(T, s)$  are defined in terms of quantized energy

levels for the reactants and generalized transition state. However, the reaction-coordinate motion is still treated classically. Analogous to the treatment of atom-diatom reactions,<sup>2,3</sup> quantal effects on the reaction-coordinate motion can be included by multiplying  $k^{\text{CVT}}(T)$  by a semiclassical transmission coefficient  $\kappa^{\text{CVT/SAG}}$  based on the vibrationally adiabatic ground-state potential energy curve. This curve is defined by

$$V_a^{\text{G}}(s) = V_{\text{MEP}}(s) + \epsilon_{\text{int}}^{\text{G}}(s), \quad (8)$$

where  $\epsilon_{\text{int}}^{\text{G}}(s)$  is the zero-point energy in the generalized transition state at  $s$ . Then<sup>3</sup>

$$\kappa^{\text{CVT/SAG}}(T) = \frac{\int_0^\infty P^{\text{SAG}}(E) \exp(-\beta E) dE}{\int_{V_a^{\text{G}}[s_{\text{CVT}}^{\text{GT}}(T)]}^\infty \exp(-\beta E) dE}, \quad (9)$$

where  $E$  is the total energy with respect to the overall zero of energy (equilibrium potential energy of infinitely separated reactants) and  $P^{\text{SAG}}(E)$  is a semiclassical approximation to the one-dimensional transmission probability through a (modified or unmodified) potential energy curve  $V_a^{\text{G}}(s)$ . The possible modifications to  $V_a^{\text{G}}(s)$  and the other complications in calculating  $P^{\text{SAG}}(E)$  arise from attempts to include some account of reaction-coordinate curvature or of reaction-coordinate nonseparability or both.<sup>3,8,14,19-22</sup> The corrected CVT rate constant is given by<sup>3</sup>

$$k^{\text{CVT/SAG}}(T) = k^{\text{CVT}}(T) \kappa^{\text{CVT/SAG}}(T). \quad (10)$$

Conventional transition-state theory (TST) is a special case of GTST with  $s = 0$ . The TST rate constant is

$$k^{\ddagger}(T) = k^{\text{GT}}(T, s = 0). \quad (11)$$

A popular procedure is to correct this rate constant for quantal effects on reaction-coordinate motion to lowest order in  $\hbar$  by the Wigner correction<sup>23</sup>

$$k^{\ddagger/\text{W}}(T) = k^{\ddagger}(T) \kappa^{\text{W}}(T), \quad (12)$$

where

$$\kappa^{\text{W}}(T) = 1 + \frac{1}{24} \left| \frac{\hbar \omega^{\ddagger}}{k_{\text{B}}T} \right|^2, \quad (13)$$

$$\omega^{\ddagger} = \left( \mu^{-1} \frac{d^2 V_{\text{MEP}}}{ds^2} \right)^{1/2}. \quad (14)$$

However, the assumptions behind this correction factor are seldom valid at the temperatures of interest.<sup>20,24</sup> A better way to correct  $k^{\ddagger}(T)$  is by the vibrationally adiabatic method. The consistent application of this approach yields<sup>3</sup>

$$k^{\ddagger/\text{SAG}}(T) = k^{\ddagger}(T) \kappa^{\ddagger/\text{SAG}}(T), \quad (15)$$

where<sup>3</sup>

$$\begin{aligned} \kappa^{\ddagger/\text{SAG}}(T) &= \kappa^{\text{CVT/SAG}} \\ &\times \exp\{\beta[V_a^{\text{G}}(s=0) - V_a^{\text{G}}[s_{\text{CVT}}^{\text{GT}}(T)]]\}, \quad (16a) \\ &= \beta \int_0^\infty P^{\text{SAG}}(E) \exp\{-\beta[E - V_a^{\text{G}}(s=0)]\} dE. \quad (16b) \end{aligned}$$

It is shown elsewhere<sup>2,3</sup> that  $k^{\ddagger/\text{SAG}}(T)$  includes some of the same effects as included in  $k^{\text{CVT}}(T)$  by variationally optimizing the dividing surface location. Thus,  $k^{\ddagger/\text{SAG}}(T)$  is a form of variational transition-state theory.

In the next section we discuss practical procedures

for the evaluation of the quantities appearing in this section.

### III. PROCEDURES

#### A. Choice of $D$

As mentioned in Sec. II, the zero of energy is the reactant equilibrium potential energy. We choose  $D$  on this scale to be the lowest classical dissociation energy of the entire system. Thus,

$$D = \min(D^A, D^B, D^C + \Delta V, D^D + \Delta V),$$

where  $D^A$  is the lowest bond dissociation energy of A, and so forth for  $D^B$ , etc., and  $\Delta V$  is the classical endoergicity of the reaction. This is consistent with our choice of an upper energy limit in our treatment of atom-diatom collisions.<sup>17</sup>

#### B. Reactant partition functions

The reactant partition function per unit volume factors as

$$\Phi^R(T) = \Phi_{\text{rel}}^{A,B}(T) Q^A(T) Q^B(T), \quad (17)$$

where the translational part is

$$\Phi_{\text{rel}}^{A,B}(T) = (2\pi\mu/\beta h^2)^{3/2} \quad (18)$$

and the internal parts  $Q^A(T)$  and  $Q^B(T)$  depend on whether A and B are atoms or molecules. It will be sufficient to consider A.

For an atom

$$Q^A(T) = Q_{\text{el}}^A(T), \quad (19)$$

where  $Q_{\text{el}}^A(T)$  is the electronic partition function given by

$$Q_{\text{el}}^A(T) = \sum_{\alpha} d_{\alpha}^A \exp[-\beta \epsilon_{\alpha}^A(\alpha)] \quad (20)$$

in terms of degeneracies  $d_{\alpha}^A$  and term values  $\epsilon_{\alpha}^A(\alpha)$ , where  $\epsilon_{\alpha}^A(\alpha=0)=0$ . Since the energy spacing between electronic levels is generally quite large, we need include only a few terms to converge the sum in Eq. (20).

For molecules, we use the separable approximation

$$Q^A(T) = Q_{\text{el}}^A(T) Q_{\text{vib}}^A(T) Q_{\text{rot}}^A(T), \quad (21)$$

where  $Q_{\text{el}}^A(T)$  is based on Eq. (20),  $Q_{\text{vib}}^A(T)$  is based on the vibrational levels  $\epsilon_m^A(n)$  of the ground electronic state, and  $Q_{\text{rot}}^A(T)$  is based on the equilibrium moments of inertia. The rotational partition function is evaluated by the standard classical approximations.<sup>25</sup> Recall that the symmetry numbers are included in  $\sigma$ , not in  $Q_{\text{rot}}^A(T)$ .<sup>18</sup> For the vibrational partition functions we have used a separable approximation whereby

$$Q_{\text{vib}}^A(T) = \sum_{m=1}^{M^A} Q_m^A(T), \quad (22)$$

where  $M^A$  is the number of bound vibrational modes in the species and  $Q_m^A(T)$  is the vibrational partition function for the  $m$ th mode

$$Q_m^A(T) = \sum_n \exp[-\beta \epsilon_m^A(n)]. \quad (23)$$

In practice, the summation in Eq. (23) terminated with

the last term for which  $\epsilon_m^A(n) < D$ . The separable vibrational energies  $\epsilon_m^A(n)$  are evaluated by performing a normal mode analysis and calculating the principal harmonic, cubic, and in some cases quartic force constants along each normal mode. These force constants are used to predict energy states  $\epsilon_m^A(n)$  up to the energy  $D$  by the methods discussed in the Appendix.

#### C. Calculation of generalized transition state quantities

The procedures described here are for the case of nonlinear geometries along the MEP. Using the R coordinate system defined below Eq. (2), we find the saddle point by a Newton-Raphson search<sup>26</sup> in  $3N-6$  dimensions for the solution of  $\nabla V=0$ , where  $V$  is the potential energy surface. Then we calculate the  $3N \times 3N$  force constant matrix  $\mathbf{F}$ , where

$$F_{i\gamma, i'\gamma'} = \frac{\partial^2 V}{\partial x_{i\gamma} \partial x_{i'\gamma'}}. \quad (24)$$

$\mathbf{F}$  is diagonalized, yielding six zero eigenvalues, one negative eigenvalue, and  $3N-7$  positive eigenvalues. The frequency for the negative eigenvalue is  $\omega^\ddagger$ , and the corresponding normalized eigenvector is called  $\mathbf{L}_{3N}$ . Starting at the saddle point, we compute geometries  $\mathbf{x}(s)$  along the MEP by taking steps of length  $\Delta s$ . We take the first step in the direction  $\mathbf{L}_{3N}$  or  $-\mathbf{L}_{3N}$  and subsequent steps in the direction of the negative gradient. To attempt to reduce the numerical error in this algorithm, we included an option in the program to periodically refine the MEP by a modified version of the method of Ishida *et al.*<sup>27</sup>

To obtain  $Q^{\text{GT}}(T, s)$ , a mode analysis is performed at regular intervals along the MEP. As discussed by Miller *et al.*,<sup>8</sup> for  $s \neq 0$  the modes orthogonal to the MEP are obtained from the projected force constant matrix

$$\mathbf{F}^{\text{P}}(s) = [\mathbf{1} - \mathbf{P}(s)] \mathbf{F}(s) [\mathbf{1} - \mathbf{P}(s)], \quad (25)$$

where  $\mathbf{F}(s)$  is the matrix of Eq. (24) evaluated on the MEP at  $\mathbf{x}(s)$  and the projector  $\mathbf{P}$  is used to project out the reaction coordinate and the six coordinates corresponding to overall rotations and translations

$$P_{i\gamma, i'\gamma'}(s) = v_{i\gamma}(s) v_{i'\gamma'}(s) + (m_A + m_B)^{-1} (m_i m_{i'})^{1/2} \delta_{\gamma\gamma'} \\ + \sum_{\alpha\beta\gamma\gamma' \neq x, y, z} \epsilon_{\alpha\beta\gamma} x_{i\beta}(s) [\mathbf{I}_0^{\text{A}}(s)^{-1}]_{\alpha\alpha'} \epsilon_{\alpha'\beta'\gamma'} x_{i'\beta'}(s), \quad (26)$$

where

$$v_{i\gamma}(s) = \{-|\nabla V[\mathbf{x}(s)]|^{-1} \nabla V[\mathbf{x}(s)]\}_{i\gamma}, \quad (27)$$

$\delta_{\gamma\gamma'}$  is the Kronecker delta,  $\epsilon_{\alpha\beta\gamma}$  is the totally asymmetric Cartesian tensor,<sup>28</sup> and  $\mathbf{I}_0^{\text{A}}(s)$  is the moment of inertia tensor in  $\mathbf{x}$  coordinates at  $\mathbf{x}(s)$ . Diagonalization of  $\mathbf{F}^{\text{P}}(s)$  gives seven zeros and  $3N-7$  positive eigenvalues corresponding to the bound modes of the generalized transition state; the matrix of corresponding normalized eigenvectors will be called  $\mathbf{L}^{\text{GT}}(s)$ . These eigenvectors define a set of generalized normal modes for each  $s$ . The second, third, and in some cases fourth derivatives of  $V$  are calculated along each bound-mode direction given by a column of  $\mathbf{L}^{\text{GT}}(s)$ . These are used to approxi-

mate the quantized energies  $\epsilon_m(n, s)$  up to energy  $D - V_{\text{MEP}}(s)$  by a Morse model, a mixed quadratic-quartic model, or a modified version of Dunham's semiclassical method. These methods are discussed in detail in the Appendix. From the separable zero-point energies we calculate the total zero-point energy by

$$\epsilon_{\text{int}}^G(s) = \sum_m \epsilon_m(n=0, s), \quad (28)$$

the vibrationally adiabatic ground-state potential curve  $V_a^G(s)$  by Eq. (8), and the vibrational partition function  $Q_m^{GT}(T)$  for mode  $m$  by summation over vibrational quantum numbers  $n$  up to energy  $D - V_{\text{MEP}}(s)$ . The generalized transition-state partition function is calculated by

$$Q^{GT}(T, s) = Q_{\text{el}}^{GT}(T, s) Q_{\text{vib}}^{GT}(T, s) Q_{\text{rot}}^{GT}(T, s). \quad (29)$$

$Q_{\text{el}}^{GT}(T, s)$  is calculated by the summation

$$Q_{\text{el}}^{GT}(T, s) = \sum_{\alpha} d_{\alpha}^{GT}(s) \exp\{-\beta[\epsilon_{\text{el}}^{GT}(\alpha, s) - V_{\text{MEP}}(s)]\} \quad (30)$$

in terms of degeneracies  $d_{\alpha}^{GT}(s)$  and electronic energies  $\epsilon_{\text{el}}^{GT}(\alpha, s)$ , where  $\epsilon_{\text{el}}^{GT}(\alpha=0, s) = V_{\text{MEP}}(s)$ .  $Q_{\text{vib}}^{GT}(T, s)$  is calculated by

$$Q_{\text{vib}}^{GT}(T, s) = \prod_{m=1}^{3N-7} Q_m^{GT}(T, s) \quad (31)$$

and  $Q_{\text{rot}}^{GT}(T, s)$  is approximated classically. Finally, the generalized standard-state free energy of activation at  $s$  is calculated by

$$\Delta G^{GT,0}(T, s) = V_{\text{MEP}}(s) - \beta^{-1} \ln \frac{Q^{GT}(T, s)}{\Phi R(T) K^0}. \quad (32)$$

At each temperature, a quartic fit near the maximum is used to interpolate to the maximum of  $\Delta G^{GT,0}(T, s)$ , yielding  $s^{CVT}(T)$  and hence, by Eqs. (6b) and (7b),  $k^{CVT}(T)$ .

## D. Calculation of transmission coefficients

For atom-diatom collisions it has been found necessary to include the curvature of the MEP in order to accurately model the quantal effects on reaction-coordinate motion. There are several ways in which this could be done<sup>3,8,14,19-22</sup> but in this article we consider only two, the small-curvature method<sup>22</sup> (SCSAG) and an approximate second-order method based on the work of Gray *et al.*<sup>21</sup>

In the small-curvature method one calculates an effective reduced mass  $\mu_{\text{eff}}(s)$  to account for reaction-coordinate curvature at a displacement  $s$  from the saddle point. This effective reduced mass is given by<sup>22</sup>

$$\mu_{\text{eff}}^{SC}(s) = \mu \prod_{m=1}^{3N-7} \min \left\{ 1, \frac{\exp[-2a_m - a_m^2 + (dt_m/ds)^2]}{1} \right\}, \quad (33)$$

where

$$a_m(s) = |B_{m,3N}(s)t_m(s)|. \quad (34)$$

$t_m(s)$  is the distance in  $\mathbf{x}$  space to the classical turning point of the zero-point motion of bound mode  $m$  on the concave side of the MEP at  $s$ , and  $B_{m,3N}(s)$  denotes the component of the curvature along the direction of bound mode  $m$  at  $s$ . This component is given by<sup>8</sup>

$$B_{m,3N}(s) = -[\text{sign}(s)] \sum_i \sum_{\gamma} \frac{dv_{i\gamma}}{ds} L_{i\gamma,m}^{GT}(s), \quad (35)$$

where mode  $3N$  is the reaction coordinate. Finally,  $t_m(s)$  is calculated by the methods discussed in the Appendix. Having obtained  $\mu_{\text{eff}}(s)$ , one then approximates  $P^{\text{SAG}}(E)$  by a one-dimensional uniform semiclassical calculation using the potential  $V_a^G(s)$  and the reduced mass  $\mu_{\text{eff}}^{SC}(s)$ .<sup>22,29</sup>

Gray *et al.* have derived a transmission probability for tunneling in terms of the second-order action-angle variables in the interaction region.<sup>21</sup> Their result involves a correction  $A$  to the vibrationally adiabatic ground-state potential curve, and it also involves an effective-masslike term  $B$  and a higher-order term  $C$  that account for reaction-path curvature, changes in vibrational frequencies as functions of  $s$ , and nonadiabatic couplings of the generalized normal modes. (The  $A$  term involves only the latter two of these three effects.) Here we restrict ourselves to vibrationally adiabatic tunneling through  $V_a^G(s)$  with the assumption of separability for the generalized normal modes. Thus, we neglect the correction  $A$  to  $V_a^G(s)$  and we neglect the nonadiabatic mode-mode couplings  $B_{mm'}$ . [In separate calculations we determined that the contributions of the vibrational-frequency-change terms to  $A$  are generally small ( $< 10^{-4}$  a.u.) for the cases considered here, and the effect of these terms on the calculated rate constants would be less than 5% for all temperatures considered in this article. The nonadiabatic mode-mode couplings are numerically inconvenient to calculate, and we have not estimated their magnitude for the cases studied here. We note in this regard that one of the advantages of the SCSAG formalism is that the Coriolis terms do not contribute<sup>22</sup>; hence, they need not be neglected as a separate approximation. We also note that tests for atom-diatom collisions have shown that the small-curvature method is more reliable than the second-order one.<sup>30</sup> Nevertheless, we include the approximate second-order calculations in the present study in order to show the differences in the predictions when two different methods are used to treat the effect of reaction-coordinate curvature on the tunneling.] Under the assumptions mentioned above, the expression for the imaginary reaction-coordinate momentum at total energy  $E$  in the tunneling region may be written<sup>21</sup>

$$\text{Im } p_s^{\text{SOSAG}}(s) = \{2\mu[V_a^G(s) - E]\}^{1/2} \{B^A(s) + \frac{1}{2} + \{[B^A(s) + \frac{1}{2}]^2 + 4C(s)[V_a^G(s) - E]\}^{1/2}\}^{-1/2}, \quad (36)$$

where

$$B^A(s) = \frac{3}{4} \sum_m [a_m(s)]^2 - \frac{1}{32\mu} \sum_m \frac{[d\epsilon_m^{GT}(n=0, s)/ds]^2}{[\epsilon_m^{GT}(n=0, s)]^3} \quad (37)$$

and

$$C(s) = \frac{1}{4} \sum_m \frac{[a_m(s)]^2}{\epsilon_m^{GT}(n=0, s)}. \quad (38)$$

The terms in Eqs. (37) and (38) reduce to those derived by Gray *et al.* using a harmonic assumption for the potential energy along each mode, but we have written their result in terms of quantities that are defined

for both the harmonic and anharmonic case. This provides a nonrigorous, nonunique anharmonic generalization of their result. The superscript SOSAG in Eq. (36) stands for the second-order semiclassical adiabatic ground-state transmission coefficient, which we will use as a convenient abbreviation for the present approximation to the second-order method of Gray *et al.*<sup>21</sup> Equation (36) is to be substituted for the neglect-of-curvature approximation (minimum-energy-path approximation<sup>3</sup>)

$$\text{Im } p_s^{\text{MEP}}(s) = \{2\mu[V_a^G(s) - E]\}^{1/2} \quad (39)$$

in the uniform semiclassical calculations. It is interesting to compare the SOSAG result and the SCSAG result in more detail. For small curvature, Eq. (36) reduces to

$$\text{Im } p_s^{\text{SOSAG}}(s) \underset{B_{m,3N} \rightarrow 0}{\sim} \text{Im } p_s^{\text{MEP}}(s) / [1 + 2B^A(s) + \dots]^{1/2}. \quad (40)$$

Comparing Eqs. (39) and (40) shows that, in the small-curvature limit, the SOSAG method may also be cast as an effective-reduced-mass method. In particular,

$$\mu_{\text{eff}}^{\text{SOSAG}}(s) \underset{B_{m,3N} \rightarrow 0}{\sim} \mu \left\{ 1 - \frac{3}{2} \sum_m [a_m(s)]^2 - [V_a^G(s) - E] \times \sum_m \frac{[a_m(s)]^2}{\epsilon_m^{\text{GT}}(n=0, s)} + \frac{1}{16\mu} \sum_m \frac{[d\epsilon_m^{\text{GT}}(n=0, s)/ds]^2}{[\epsilon_m^{\text{GT}}(n=0, s)]^3} \right\}. \quad (41)$$

In contrast, Eq. (33) yields

$$\mu_{\text{eff}}^{\text{SC}}(s) \underset{B_{m,3N} \rightarrow 0}{\sim} \mu \left\{ 1 + \sum_m [-2a_m(s) + a_m^2(s) + (dt_m/ds)^2] \right\}. \quad (42)$$

The existence of terms linear in the curvature in Eq. (42) but not in Eq. (41) shows that the SCSAG method predicts more tunneling than the SOSAG method; and it is because the SCSAG method predicts more tunneling that it has been found empirically<sup>30</sup> to be a more reliable theory for atom-diatom reactions.

The methods used for the quadratures in the semiclassical calculation are the same as used in our previous work.<sup>3,23</sup>

## E. Computational details

For OH + H<sub>2</sub>, we optimized the numerical parameters so that the resulting numerical uncertainties in the calculated rates are as small as possible for the amount of computational effort involved and in any case are less than 1%. For this accuracy, we followed the analytic gradient with a step size of 0.0001a<sub>0</sub> and omitted the refinement of Ishida *et al.*,<sup>27</sup> which actually slowed convergence for such small step sizes. Every 0.01a<sub>0</sub> along the MEP, we calculated the matrix (24) by a two-point central difference of the analytic gradient

$$F_{i\gamma, i'\gamma'} = \frac{1}{2\delta} \{ \nabla V[\mathbf{x}(s) + \delta \mathbf{e}_{i\gamma}] - \nabla V[\mathbf{x}(s) - \delta \mathbf{e}_{i\gamma}] \}_{i'\gamma'}, \quad (43)$$

where  $\mathbf{e}_{i\gamma}$  is the unit vector along the  $x_{i\gamma}$  direction and  $\delta$

is a small parameter. At these locations we performed a mode analysis using Eqs. (25)–(27) and, for use in anharmonicity calculations, we calculated the directional third and fourth derivatives of the potential along the bound-mode directions using three- and four-point central-difference numerical derivatives of the analytic gradient along the normal mode directions, i.e.,

$$k_{mmm}(s) = \frac{1}{6\delta^2} \{ \nabla V[\mathbf{x}(s) + \delta \mathbf{L}_m^{\text{GT}}(s)] + \nabla V[\mathbf{x}(s) - \delta \mathbf{L}_m^{\text{GT}}(s)] - 2\nabla V[\mathbf{x}(s)] \} \cdot \mathbf{L}_m^{\text{GT}}(s), \quad (44)$$

$$k_{mmmm}(s) = \frac{1}{48\delta^3} \{ \nabla V[\mathbf{x}(s) + 2\delta \mathbf{L}_m^{\text{GT}}(s)] - 2\nabla V[\mathbf{x}(s) + \delta \mathbf{L}_m^{\text{GT}}(s)] + 2\nabla V[\mathbf{x}(s) - \delta \mathbf{L}_m^{\text{GT}}(s)] - \nabla V[\mathbf{x}(s) - 2\delta \mathbf{L}_m^{\text{GT}}(s)] \} \cdot \mathbf{L}_m^{\text{GT}}(s). \quad (45)$$

A step size  $\delta$  of 0.0001a<sub>0</sub> for Eqs. (43)–(45) was found to provide converged results. From these derivatives, vibrational energy levels were computed by the methods discussed in the Appendix, thereby allowing the calculation of vibrational partition functions, and hence  $\Delta G^{\text{GT},0}(T, s)$ .

The curvature components of Eq. (35) were computed at each location at which a mode analysis was performed, using one-sided numerical differencing of the gradient with a step size of 0.0001a<sub>0</sub> to obtain converged results. Concave-side turning points were then obtained by the methods discussed in the Appendix. Since the sign of  $\mathbf{L}_m^{\text{GT}}(s)$ , and hence those of  $B_{m,3N}$  and  $t_m$ , can change arbitrarily along the reaction path, the turning point derivative term  $(dt_m/ds)^2$  was obtained by a two-point central difference of the absolute values of the turning point with a step size of 0.01a<sub>0</sub>. For the first ( $s_1$ ) and last ( $s_M$ ) points of the path from the saddle point at which a mode analysis was performed, the turning point derivative term was obtained by Lagrangian interpolation<sup>31</sup>

$$[dt_m(s_1)/ds]^2 = \{ [-3|t_m(s_1)| + 4|t_m(s_2)| - |t_m(s_3)|] / (s_3 - s_1) \}^2, \quad (46)$$

$$[dt_m(s_M)/ds]^2 = \{ [|t_m(s_{M-2})| - 4|t_m(s_{M-1})| + 3|t_m(s_M)|] / (s_M - s_{M-2}) \}^2. \quad (47)$$

The zero-point energy derivative term appearing in Eq. (37) was also computed in a similar fashion. Before taking derivatives, the values of  $t_m(s)$  and  $\epsilon_m^{\text{GT}}(n=0, s)$  must be arranged in the same order of the modes for each  $s$  value; this was done by using the overlaps of the vectors in  $\mathbf{L}_m^{\text{GT}}(s_i)$  with those in  $\mathbf{L}_m^{\text{GT}}(s_{i+1})$ . Since the curvature components cannot be determined at the saddle point from Eq. (35), we used linear interpolation between the two closest points for  $\mu_{\text{eff}}^{\text{SC}}$ ,  $B^A$ , and  $C$  at  $s=0$ .

A further complication in the calculation of transmission coefficients for the OH + H<sub>2</sub> system results from a local minimum in  $V_{\text{MEP}}(s)$  on the reactant side of the saddle point at  $s = -1.77a_0$ . Since the algorithm we used to find the MEP could not be applied beyond this minimum, and since the region beyond the minimum was found to be important only for very low-energy tunneling probabilities, we obtained  $V_a^G(s)$ ,  $\mu_{\text{eff}}^{\text{SC}}(s)$ ,  $B^A(s)$ , and  $C(s)$  in this region by the following extrapolation:

$$V_a^G(s) = V_a^G(s = -\infty) + [V_a^G(s = s_{\text{min}}) - V_a^G(s = -\infty)] \exp[\alpha(s - s_{\text{min}})], \quad (48)$$

and similar expressions for the other quantities. For the calculations presented here, we used  $\alpha = 0.75a_0^{-1}$ ; varying  $\alpha$  over the reasonable range from 0.5 to  $1.0a_0^{-1}$  caused less than a 2% difference in the calculated rate constants for all temperatures in this study.

The only quantities needed in our calculations that were not computed from the potential energy surface of Ref. 9 are the electronic energies and degeneracies of the reactants, products, and generalized transition states. The values of these parameters<sup>32</sup> used in the present study are listed in Table I.

#### IV. RATE CONSTANTS

Table II gives conventional transition-state theory results (for which our abbreviations are ‡ for unit transmission coefficient and ‡/W for inclusion of the Wigner<sup>23</sup> transmission coefficient) for the original *ab initio* saddle point and reactant properties<sup>33</sup> and also for the saddle point and reactant properties computed from the fitted surface of Schatz and Elgersma.<sup>9</sup> The results in Table II are based on the harmonic (H) approximation so we denote them H‡ and H‡/W. For the original *ab initio* surface there is insufficient data to include anharmonicity or to calculate  $k^{\text{GT}}(T, s)$  for any  $s$  except  $s=0$ . Thus, the comparison of the *ab initio* and fitted surfaces can only be carried out within the H‡ and H‡/W approximations. Table II shows that rate constants obtained from the fitted surface<sup>9</sup> are somewhat different from those obtained using the parameters of the original *ab initio* surface over the whole temperature range considered. These differences are mainly due to differences in the moments of inertia caused by different saddle point geometries. Table II also shows that inclusion of spin-orbit coupling in OH(<sup>2</sup>Π) makes a significant difference in the rate constant, especially at low temperature. We therefore included spin-orbit coupling in all of the calculations discussed below.

Table III gives both ‡ and CVT results for three different methods of including anharmonicity, as discussed in detail in the Appendix. In particular, the anharmonicity algorithms used for the results in the table are Morse-quadratic-quartic schemes I and III and the cor-

TABLE II. Calculated rate constants ( $\text{cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$ ) for Reaction (R1).

| T(K) | <i>Ab initio</i> <sup>a</sup> |           | Fitted <sup>b</sup> |           |
|------|-------------------------------|-----------|---------------------|-----------|
|      | H‡                            | H‡/W      | H‡                  | H‡/W      |
| 200  | 2.69(−17) <sup>c,d</sup>      | 1.86(−16) | 2.08(−17)           | 1.25(−16) |
|      | 1.84(−17) <sup>d</sup>        | 1.27(−16) | 1.42(−17)           | 8.56(−17) |
| 300  | 2.53(−15)                     | 9.16(−15) | 1.86(−15)           | 6.02(−15) |
|      | 1.91(−15)                     | 6.92(−15) | 1.41(−15)           | 4.55(−15) |
| 400  | 2.47(−14)                     | 6.11(−14) | 1.76(−14)           | 3.97(−14) |
|      | 1.98(−14)                     | 4.90(−14) | 1.41(−14)           | 3.18(−14) |
| 600  | 2.63(−13)                     | 4.35(−13) | 1.82(−13)           | 2.84(−13) |
|      | 2.25(−13)                     | 3.73(−13) | 1.56(−13)           | 2.43(−13) |
| 1000 | 2.30(−12)                     | 2.84(−12) | 1.58(−12)           | 1.90(−12) |
|      | 2.09(−12)                     | 2.58(−12) | 1.43(−12)           | 1.72(−12) |
| 1500 | 9.30(−12)                     | 1.03(−11) | 6.39(−12)           | 6.97(−12) |
|      | 8.71(−12)                     | 9.63(−12) | 5.99(−12)           | 6.53(−12) |
| 2400 | 3.90(−11)                     | 4.06(−11) | 2.69(−11)           | 2.78(−11) |
|      | 3.75(−11)                     | 3.90(−11) | 2.58(−11)           | 2.67(−11) |
| 4000 | 1.59(−10)                     | 1.61(−10) | 1.09(−10)           | 1.10(−10) |
|      | 1.55(−10)                     | 1.57(−10) | 1.06(−10)           | 1.08(−10) |

<sup>a</sup>Parameters from Ref. 33.

<sup>b</sup>Parameters from the fitted surface of Ref. 9 as computed in the present work.

<sup>c</sup>Power of 10 in parentheses.

<sup>d</sup>Upper entry includes spin-orbit coupling in OH(<sup>2</sup>Π), while lower entry neglects it, i.e., the energy of the second OH level of Table I is also taken as 0 for the lower entries.

rected Dunham  $\chi$  method. By comparing Tables II and III, it is clear that the inclusion of anharmonicity has a large effect on the calculated rate constants; at low temperature the anharmonic results differ from the harmonic ones by factors of about 2 to 4. Table III also shows that although there is little difference in the results obtained with either Morse-quadratic-quartic model except at high temperature, the corrected Dunham  $\chi$  model yields results which are substantially lower than those of the Morse-quadratic-quartic models. The differences are caused primarily by the differing predictions of Eqs. (A26) and (A27) and the Morse model for the two low-frequency in-plane bending modes of the conventional and generalized transition states.

Our choice of which method to use to treat anharmonicity in further calculations is based not on considerations of accuracy but on the ability of an anharmonicity scheme to predict continuous values of the quantities entering the tunneling calculations. Along the MEP, a given mode sometimes requires either a change from Morse approximation II to Morse approximation I or, for the Dunham  $\chi$  model, a modification in  $Y'_{20}$  (cf. the Appendix) in order for the energy levels to reach  $D$ . Thus, the Morse-quadratic-quartic scheme III and the corrected Dunham  $\chi$  model will not provide smooth concave-side turning points or zero-point energies as functions of  $s$ , leading to difficulties in the calculation of SCSAG or SOSAG transmission coefficients. For this reason, only the Morse I plus qq model was used to obtain the anharmonic results presented in the following tables. One further point worth noting is that in Morse

TABLE I. Electronic energy levels.

|                                                                   | $d_\alpha$ | $\epsilon_{e1}(\alpha)^a$<br>( $\text{cm}^{-1}$ ) |
|-------------------------------------------------------------------|------------|---------------------------------------------------|
| Reactants OH( <sup>2</sup> Π <sub>3/2</sub> )                     | 2          | 0                                                 |
| OH( <sup>2</sup> Π <sub>1/2</sub> )                               | 2          | 140 <sup>b</sup>                                  |
| H <sub>2</sub> ( <sup>1</sup> Σ <sub>g</sub> <sup>+</sup> )       | 1          | 0                                                 |
| Generalized transition states H <sub>2</sub> OH( <sup>2</sup> A') | 2          | 0                                                 |
| Products H <sub>2</sub> O( <sup>1</sup> A <sub>1</sub> )          | 1          | 0                                                 |
| H( <sup>2</sup> S)                                                | 2          | 0                                                 |

<sup>a</sup>Relative to  $V_{\text{MEP}}(s)$ .

<sup>b</sup>Reference 32.

TABLE III. Comparison of calculated rate constants ( $\text{cm}^3 \text{molecule}^{-1} \text{s}^{-1}$ ) for Reaction (R1) for the fitted potential energy surface of Ref. 9 obtained by various treatments of anharmonicity.

| T(K) | Morse-quadratic-quartic I |           | Morse-quadratic-quartic III |           | Corrected Dunham $\alpha$ |           |
|------|---------------------------|-----------|-----------------------------|-----------|---------------------------|-----------|
|      | ‡                         | CVT       | ‡                           | CVT       | ‡                         | CVT       |
| 200  | 1.11(-17)                 | 4.08(-18) | 1.10(-17)                   | 4.06(-18) | 5.03(-18)                 | 1.05(-18) |
| 298  | 1.14(-15)                 | 6.01(-16) | 1.13(-15)                   | 5.99(-16) | 6.37(-16)                 | 2.27(-16) |
| 400  | 1.26(-14)                 | 8.05(-15) | 1.25(-14)                   | 8.02(-15) | 7.56(-15)                 | 3.57(-15) |
| 600  | 1.40(-13)                 | 1.09(-13) | 1.39(-13)                   | 1.08(-13) | 8.59(-14)                 | 5.27(-14) |
| 1000 | 1.26(-12)                 | 1.11(-12) | 1.25(-12)                   | 1.10(-12) | 7.15(-13)                 | 5.15(-13) |
| 1500 | 5.14(-12)                 | 4.70(-12) | 5.05(-12)                   | 4.62(-12) | 2.56(-12)                 | 1.91(-12) |
| 2400 | 2.14(-11)                 | 1.97(-11) | 2.08(-11)                   | 1.92(-11) | 8.66(-12)                 | 6.28(-12) |
| 4000 | 8.59(-11)                 | 7.86(-11) | 8.16(-11)                   | 7.49(-11) | 2.56(-11)                 | 1.74(-11) |

approximation I the dissociation energy of the  $\text{H}_2$  reactant is, for the sake of consistency (see Sec. III A), based on  $D$  rather than  $D_e(\text{H}_2)$ . This lowers the Morse dissociation energy from the value on the fitted surface of 0.151 548 a.u.<sup>9</sup> to  $D = D_e(\text{OH}) = 0.148 201$  a.u.<sup>9</sup> This made less than 0.3% difference in the results.

In Table IV, we present various calculated rate constants for Reaction (R1). Several features of these results deserve mention: (1) Varying the location of the dividing surface has a substantial effect for this reaction, especially at low temperature. This effect arises mainly from a stronger  $\text{H}_2$  bond for the CVT bottleneck than at the saddle point. This is discussed further in Sec. V. (2) The inclusion of reaction-coordinate tunneling is very large for  $T \leq 600$  K for this reaction. In addition, the SCSAG method provides much larger transmission probabilities than does the SOSAG method, as anticipated in Sec. IIID. (3) The effect of including anharmonicity is larger for CVT rates than for conven-

tional ones, and is still larger for the rates which include tunneling. The reason for the latter fact is that anharmonicity affects the whole  $V_a^G(s)$  function, which provides the barrier for the tunneling calculations.

Table IV also compares our results to the experimental measurements of Ravishankara and co-workers.<sup>11</sup> Based on our experience with atom-diatom reactions,<sup>6,22</sup> we believe the anharmonic results in the CVT/SCSAG column to be the most reliable of the theoretical columns in Table IV. It is encouraging that the discrepancies between the experimental results and these totally *ab initio* theoretical calculations are less than a factor of 2 over the whole factor of 3.4 temperature range for which comparison is possible. In particular, the theoretical results are higher than the experimental ones by a factor of 1.7 at 298 K, they agree with experiment within experimental error at 400 and 600 K, and they are lower than the experimental results by a factor of 0.6 at 1000 K. Considering the several possible sources

TABLE IV. Calculated<sup>a</sup> and experimental rate constants ( $\text{cm}^3 \text{molecule}^{-1} \text{s}^{-1}$ ) for Reaction (R1).

| T(K) | ‡                      | ‡/W       | CVT       | CVT/SOSAG | CVT/SCSAG | Expt. <sup>b</sup> |
|------|------------------------|-----------|-----------|-----------|-----------|--------------------|
| 200  | 1.11(-17) <sup>c</sup> | 6.68(-17) | 4.08(-18) | 2.68(-16) | 1.77(-15) | ...                |
|      | 2.08(-17) <sup>d</sup> | 1.25(-16) | 9.97(-18) | 9.26(-16) | 6.87(-15) |                    |
| 298  | 1.14(-15)              | 3.72(-15) | 6.01(-16) | 3.30(-15) | 1.04(-14) | 6.08 ± 0.37(-15)   |
|      | 1.75(-15)              | 5.72(-15) | 1.11(-15) | 6.84(-15) | 2.37(-14) |                    |
| 400  | 1.26(-14)              | 2.83(-14) | 8.05(-15) | 1.97(-14) | 4.03(-14) | 3.7 ± 1.1(-14)     |
|      | 1.76(-14)              | 3.97(-14) | 1.30(-14) | 3.35(-14) | 7.26(-14) |                    |
| 600  | 1.40(-13)              | 2.18(-13) | 1.09(-13) | 1.58(-13) | 2.23(-13) | 2.9 ± 0.9(-13)     |
|      | 1.82(-13)              | 2.84(-13) | 1.57(-13) | 2.31(-13) | 3.35(-13) |                    |
| 1000 | 1.26(-12)              | 1.52(-12) | 1.11(-12) | 1.25(-12) | 1.42(-12) | 2.4 ± 0.7(-12)     |
|      | 1.58(-12)              | 1.90(-12) | 1.50(-12) | 1.66(-12) | 1.91(-12) |                    |
| 1500 | 5.14(-12)              | 5.60(-12) | 4.70(-12) | 4.90(-12) | 5.19(-12) | ...                |
|      | 6.39(-12)              | 6.97(-12) | 6.26(-12) | 6.36(-12) | 6.78(-12) |                    |
| 2400 | 2.14(-11)              | 2.21(-11) | 1.97(-11) | 2.00(-11) | 2.04(-11) | ...                |
|      | 2.69(-11)              | 2.78(-11) | 2.66(-11) | 2.63(-11) | 2.69(-11) |                    |
| 4000 | 8.59(-11)              | 8.70(-11) | 7.86(-11) | 7.89(-11) | 7.96(-11) | ...                |
|      | 1.09(-10)              | 1.10(-10) | 1.08(-10) | 1.07(-10) | 1.08(-10) |                    |

<sup>a</sup>For the fitted potential energy surface of Ref. 9.

<sup>b</sup>From Ref. 11; for  $T = 298$  K we quote the actual tabulated values, while for  $T > 298$  K we employ those authors' three-parameter fit with 30% error bars.

<sup>c</sup>Upper entry in each case includes anharmonicity in the independent-normal-coordinate representation by the Morse-quadratic-quartic scheme I as discussed in the Appendix.

<sup>d</sup>The lower entry in each case is the harmonic result.

TABLE V. Calculated and experimental average activation energies over the temperature range 298–1000 K for Reaction (R1).

| Method <sup>a</sup> | $E_a$ (kcal/mol) |
|---------------------|------------------|
| ‡                   | 5.9              |
| ‡/W                 | 5.1              |
| CVT                 | 6.3              |
| CVT/SOSAG           | 5.0              |
| CVT/SCSAG           | 4.1              |
| Experiment          | 5.0              |

<sup>a</sup>Theoretical values are based on the fitted surface of Ref. 9 and include anharmonicity by the Morse-quadratic-quartic scheme I.

of error—the *ab initio* surface, the surface fit, the CVT formulation, the treatment of anharmonicity, and the treatment of tunneling—these sizes of discrepancy are probably better than one can expect in general.

Using two-temperature Arrhenius fits at 298 and 1000 K (not because the Arrhenius plot is linear but simply to obtain an average Arrhenius slope over the interesting temperature region), we calculated average activation energies for each of the methods of Table IV. The results are compared to each other and to the experimental average activation energy for this temperature range in Table V. First of all, we note that the dynamical methods that we consider less reliable led to activation energies differing from the CVT/SCSAG one by as much as 2 kcal/mol. This would be a serious error if one used such a method to judge the reliability of a potential energy surface by comparison to experiment. Second, we discuss the actual value of the CVT/SCSAG activation energy. The saddle point height on the

Born–Oppenheimer potential energy surface [maximum of  $V_{\text{MEP}}(s)$  with respect to reactants] is 6.1 kcal/mol. The zero-point energies of reactants and of the saddle point are almost identical so this value provides a first approximation to the ‡ value of  $E_a$ , and Table V shows that it is a good approximation. The CVT calculation is based on the best bottleneck in a variational sense. The maximum of  $V_a^G(s)$  exceeds  $V_a^G(s=0)$  by 0.4 kcal/mol so 0.4 kcal/mol is a first approximation to the expected difference of the CVT and ‡ values of  $E_a$ . Again Table V bears out this kind of interpretation in terms of zero-point energies. The three methods of including tunneling lower the predicted  $E_a$  by 0.8–2.2 kcal/mol. The final prediction agrees with experiment within 0.9 kcal/mol.

Table VI compares calculated and experimental values of the kinetic isotope effect, which is the ratio of the rate constant for Reaction (R1) to that for



First of all, we see that the ‡ values are higher than the experimental ones over the whole temperature range. Any correction for tunneling would be expected to increase the predicted kinetic isotope effect, and indeed the Wigner correction does increase it. The ‡/W kinetic isotope effects exceed the experimental ones by factors of 1.5–1.25 over the experimental temperature range. The CVT kinetic isotope effect is considerably lower than the experimental one at 298–400 K and slightly higher at 600–1000 K. Thus, the CVT results are in a better position to be improved by the tunneling correction than the ‡ results were. However, the SOSAG and SCSAG methods both predict very large effects of tunneling on the kinetic isotope effect. Our own conclusion is that the quantitative disagreement of CVT/SCSAG and experimental kinetic isotope effects is probably caused primarily by deficiencies of the *ab initio* potential energy surface or the fit, but of course this is not defini-

TABLE VI. Calculated kinetic isotope effects for the fitted potential energy surface of Ref. 9.<sup>a</sup>

| T(K) | ‡                                      | ‡/W          | CVT          | CVT/SOSAG    | CVT/SCSAG    | Expt. <sup>b</sup> |
|------|----------------------------------------|--------------|--------------|--------------|--------------|--------------------|
| 200  | 4.05 <sup>c</sup><br>5.53 <sup>d</sup> | 6.60<br>9.00 | 2.10<br>3.42 | 10.0<br>20.4 | 11.1<br>24.0 | ...                |
| 298  | 3.35<br>4.09                           | 4.93<br>6.02 | 2.18<br>2.98 | 4.34<br>6.46 | 5.60<br>8.92 | 3.3 ± 0.9          |
| 400  | 2.90<br>3.34                           | 3.90<br>4.50 | 2.13<br>2.69 | 3.08<br>4.03 | 3.75<br>5.16 | 2.5 ± 0.7          |
| 600  | 2.36<br>2.58                           | 2.83<br>3.10 | 1.97<br>2.31 | 2.30<br>2.73 | 2.57<br>3.12 | 1.7 ± 0.5          |
| 1000 | 1.85<br>1.95                           | 2.00<br>2.12 | 1.70<br>1.88 | 1.78<br>1.96 | 1.86<br>2.06 | 1.6 ± 0.5          |
| 1500 | 1.60<br>1.66                           | 1.67<br>1.73 | 1.53<br>1.64 | 1.54<br>1.64 | 1.58<br>1.68 | ...                |
| 2400 | 1.45<br>1.49                           | 1.48<br>1.51 | 1.41<br>1.48 | 1.41<br>1.46 | 1.42<br>1.48 | ...                |
| 4000 | 1.38<br>1.41                           | 1.39<br>1.42 | 1.35<br>1.40 | 1.35<br>1.39 | 1.35<br>1.39 | ...                |

<sup>a</sup>See Table IV.

<sup>b</sup>From Table V of Ref. 11.

<sup>c,d</sup>See Table IV.

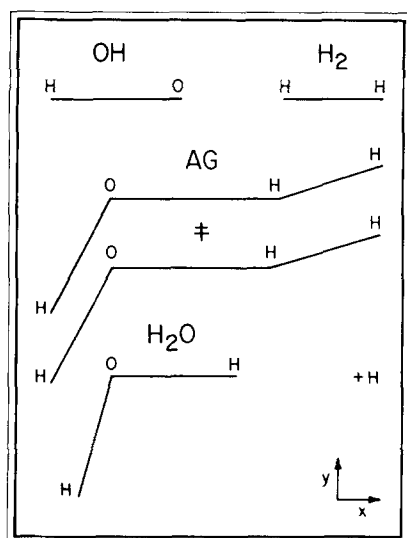


FIG. 1. Scale drawing showing the bond lengths of reactants and the geometries of the vibrationally adiabatic ground-state maximum species, the saddle-point species, and the products.

tive. There is certainly the possibility that the blame lies elsewhere, e.g., in the experiments or the tunneling correction.

## V. GENERALIZED-TRANSITION-STATE ANALYSIS

The numerical values of the rate constants and kinetic isotope effects in the preceding section provide a quantitative test of how well the Walch-Dunning-Schatz-Elgersma potential energy surface can or cannot account for the magnitudes of the observed results. They also provide a guide for extrapolation to both higher and lower temperatures than those for which experimental results are available. This kind of comparison is one

valid objective for a theoretical rate calculation. A second objective is to elucidate the character of the dynamical bottleneck and to discover those factors that are primarily responsible for the difference between the dynamical bottleneck and the conventional transition state. In this section we consider these questions.

Figure 1 shows the geometries corresponding to the adiabatic ground-state barrier maximum, denoted AG, and to the saddle point, denoted ‡. With both species positioned to have their center-of-mass at the origin, the largest difference in any Cartesian coordinate is in the  $x$  coordinate of the transferred H. This atom is  $0.14a_0$  farther from the O at the adiabatic ground-state barrier maximum than it is at the saddle point. The HOH bond angle is about  $117^\circ$  at both the adiabatic ground-state barrier maximum and the saddle point, but is  $105^\circ$  in the product  $H_2O$ . The Born-Oppenheimer potential energy curve, the vibrationally adiabatic ground-state potential energy curve, and the five vibrational frequencies for the generalized normal modes are given for these four points on the surface and for several others in Table VII. This table shows that although the geometry difference between the AG and ‡ species is small on the scale of Fig. 1, one of the vibrational frequencies ( $\omega_4$ ) is  $788\text{ cm}^{-1}$  higher at the former location. This vibration is primarily an HH stretching vibration. In this respect the location of the variationally best dividing surface for Reaction (R1) is controlled by similar factors to those controlling the atom-diatom reaction  $Cl + H_2$ . In a broader context, we note that we have found for many atom-diatom reactions with tight transition states that stretching frequencies vary strongly near the saddle points and that their dependence on  $s$  can lead to large variational corrections to conventional transition-state theory.<sup>1-3,5,34</sup> For the present case the HH stretching mode is mode 5 from  $s = -\infty$  to

TABLE VII. Variation of energetic quantities with progress along the reaction coordinate.

| $s$<br>( $a_0$ )     | $T^a$<br>(K)   | $V_{MEP}$<br>(kcal/mol) | $V_a^{G^b}$<br>(kcal/mol) | $\omega_1$<br>( $\text{cm}^{-1}$ ) | $\omega_2$<br>( $\text{cm}^{-1}$ ) | $\omega_3$<br>( $\text{cm}^{-1}$ ) | $\omega_4$<br>( $\text{cm}^{-1}$ ) | $\omega_5$<br>( $\text{cm}^{-1}$ ) |
|----------------------|----------------|-------------------------|---------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| $-\infty^c$          |                | 0.00                    | 11.10                     | 0                                  | 0                                  | 0                                  | 3620                               | 4260                               |
| -1.7                 |                | -1.27                   | 12.19                     | 355                                | 355 <sup>e</sup>                   | 488                                | 3633                               | 4172                               |
| -0.8                 |                | 0.34                    | 13.38                     | 346                                | 362 <sup>e</sup>                   | 377                                | 3589                               | 4055                               |
| -0.6                 |                | 1.62                    | 14.60                     | 397                                | 404 <sup>e</sup>                   | 425                                | 3557                               | 3933                               |
| -0.4                 |                | 3.26                    | 16.07                     | 464                                | 474 <sup>e</sup>                   | 485                                | 3523                               | 3705                               |
| -0.2                 |                | 5.01                    | 17.38                     | 523                                | 606 <sup>e</sup>                   | 612                                | 3176                               | 3528                               |
| -0.1043 <sup>c</sup> | 0 <sup>d</sup> | 5.71                    | 17.63                     | 547                                | 707 <sup>e</sup>                   | 721                                | 2699                               | 3530                               |
| -0.0982              | 300            | 5.75                    | 17.63                     | 548                                | 715 <sup>e</sup>                   | 729                                | 2659                               | 3530                               |
| -0.0842              | 600            | 5.83                    | 17.62                     | 552                                | 732 <sup>e</sup>                   | 748                                | 2563                               | 3531                               |
| -0.0702              | 1000           | 5.90                    | 17.59                     | 555                                | 750 <sup>e</sup>                   | 767                                | 2462                               | 3533                               |
| -0.0662              | 2400           | 5.92                    | 17.58                     | 556                                | 755 <sup>e</sup>                   | 773                                | 2433                               | 3534                               |
| 0.0 <sup>c</sup>     |                | 6.09                    | 17.21                     | 572                                | 830 <sup>e</sup>                   | 857                                | 1921                               | 3542                               |
| 0.1                  |                | 5.41                    | 15.96                     | 586                                | 854 <sup>e</sup>                   | 910                                | 1445                               | 3573                               |
| 0.2                  |                | 2.98                    | 13.37                     | 551                                | 757 <sup>e</sup>                   | 911                                | 1424                               | 3621                               |
| 0.3                  |                | -0.48                   | 10.24                     | 417                                | 532 <sup>e</sup>                   | 965                                | 1825                               | 3707                               |
| $\infty^c$           |                | -15.20                  | -1.60                     | 0                                  | 0                                  | 1687                               | 3865                               | 3975                               |

<sup>a</sup>If a temperature is given in this column, the row is the CVT transition state for that temperature. Other rows are given just to show the variation of these quantities with  $s$ .

<sup>b</sup>Morse-quadratic-quartic scheme I.

<sup>c</sup>Geometry shown in Fig. 1.

<sup>d</sup>The CVT transition state at 0 K is located at the vibrationally adiabatic ground-state barrier maximum.

<sup>e</sup>Out-of-plane bend. Other vibrations are in plane.

$\cong -0.31a_0$ ; after that it is mode 4. Thus, the HH stretching frequency decreases from  $4260\text{ cm}^{-1}$  at reactants to  $2433\text{--}2699\text{ cm}^{-1}$  for the various variational transition states to  $1921\text{ cm}^{-1}$  at the saddle point. In contrast, the OH stretching frequency varies only over the range  $3620\text{--}3542\text{ cm}^{-1}$  all the way from reactants to the saddle point; the spectator nature of this mode reinforces the analogy to  $\text{Cl} + \text{H}_2$ , where such a mode is absent.

To quantitatively compare the role of vibrational vs rotational effects in the variational optimization of the CVT transition state, we factor both the CVT and  $\ddagger$  rate expressions into factors associated with  $V_{\text{MEP}}(s)$ , with vibrational energies, and with rotation. Then we ratio each factor separately. The results are given in Table VIII. This table shows that the rotational effects on the variational optimization are only 3%–4% over the whole temperature range studied. This result is again similar to what we found for atom–diatom reactions with tight saddle points.<sup>1–3,5,34</sup>

Finally, we consider the approximation of Eqs. (15) and (16). We mentioned in Sec. II that  $k^{\ddagger/\text{SAG}}$  calculations provide a simplified form of variational transition-state theory. We have shown in previous applications to collinear and three-dimensional atom–diatom reactions that  $\ddagger/\text{SAG}$ -type approximations often agree remarkably well with full CVT/SAG calculations, especially at low and medium temperatures.<sup>2–4,37</sup> Since  $\ddagger/\text{SAG}$  calculations are easier than full variational transition-state theory calculations, it is interesting to examine whether this approximation retains its usefulness for polyatomic reactions. Table IX provides a test of this question for the case where the SCSAG method is used for the SAG transmission coefficient. The table shows that the use of a consistent formulation of the vibrationally adiabatic ground-state transmission coefficient again provides a very good approximation to the full variational transition-state theory calculation, and in the present case it actually yields accurate rate constants up to quite high temperature. Although one would not want to trust the  $\ddagger/\text{SAG}$  procedure when more theoretically justifiable calculations are available, it certainly provides a useful algorithm for fast estimates.

## VI. CONCLUDING REMARKS

We have applied variational transition-state theory to a reaction of a polyatomic species using a fit to an

TABLE VIII. Factor analysis of CVT transition state as compared to the conventional transition state.<sup>a</sup>

| T(K) | Factors in $k^{\ddagger}/k^{\text{CVT}}$ |            |           | $k^{\ddagger}/k^{\text{CVT}}$ |
|------|------------------------------------------|------------|-----------|-------------------------------|
|      | Potential                                | Vibrations | Rotations |                               |
| 200  | 0.400                                    | 7.09       | 0.960     | 2.72                          |
| 300  | 0.560                                    | 3.50       | 0.961     | 1.89                          |
| 600  | 0.800                                    | 1.66       | 0.967     | 1.29                          |
| 1000 | 0.907                                    | 1.29       | 0.973     | 1.14                          |
| 2400 | 0.964                                    | 1.15       | 0.974     | 1.08                          |

<sup>a</sup>Morse-quadratic-quartic scheme I.

TABLE IX. Calculated rate constants for Reaction (R1).<sup>a</sup>

| T(K) | $\ddagger$ | $\ddagger/\text{SCSAG}$ | CVT/SCSAG |
|------|------------|-------------------------|-----------|
| 200  | 1.11(–17)  | 1.69(–15)               | 1.77(–15) |
| 350  | 4.60(–15)  | 2.05(–14)               | 2.20(–14) |
| 600  | 1.40(–13)  | 2.05(–13)               | 2.23(–13) |
| 1500 | 5.14(–12)  | 5.04(–12)               | 5.19(–12) |
| 2400 | 2.14(–11)  | 2.05(–11)               | 2.04(–11) |

<sup>a</sup>Morse-quadratic-quartic scheme I.

*ab initio* potential energy surface. The results agree with experiment within a factor of 1.7 over the whole experimental temperature range 298–1000 K. This is very encouraging. Now that the practical techniques have been carried through once, we look forward to applications to other reactions, improved treatments of anharmonicity, and more systematic study of bottleneck determinants in polyatomic reaction dynamics.

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## APPENDIX

In this Appendix, we present the details of the computation of the vibrational energy levels  $\epsilon(n)$  and the concave-side ground-state turning points  $t_m$  for the various methods discussed in the text. Under the separable-mode approximation, it is sufficient to discuss only a single mode, denoted by  $m$ . Given the second, third, fourth, etc. derivatives of the energy along mode  $m$ , the potential energy for this one-dimensional oscillator can be written as

$$V(u) = \frac{1}{2}k_{mm}u^2 + k_{mmm}u^3 + k_{mmmm}u^4 + \dots, \quad (\text{A1})$$

where the corresponding harmonic frequency is

$$\omega_e = (k_{mm}/\mu)^{1/2}. \quad (\text{A2})$$

### I. Harmonic model (H)

In the harmonic model only the first term on the right-hand side of Eq. (A1) is retained. Energy levels are given by

$$\epsilon(n) = \hbar\omega_e(n + \frac{1}{2}). \quad (\text{A3})$$

The zero-point energy turning point on the concave side of the MEP is then given by

$$t_m = \pm \left[ \frac{2\epsilon(0)}{k_{mm}} \right]^{1/2} = \pm \left( \frac{\hbar}{\mu\omega_e} \right)^{1/2}, \quad (\text{A4})$$

where the sign of  $t_m$  is chosen to be opposite that of  $B_{m,3N}$ .

## II. Morse-quadratic-quartic models (Mqql, MqqII, and MqqIII)

In the Morse-quadratic-quartic models, we use different methods to calculate the energy levels, depending upon whether or not  $k_{mmm}$  is zero. If it is, we use a mixed quadratic-quartic model; if not, we use a Morse model.

### A. Mixed quadratic-quartic model

In this case the potential of Eq. (A1) is replaced by

$$V(u) = \frac{1}{2}k_{mm}u^2 + k_{mmm}u^4, \quad (\text{A5})$$

where we assume that  $k_{mmm}$  is positive. The calculation of accurate energy levels for this potential is discussed elsewhere.<sup>35</sup> The concave-side turning point for the zero-point level is given by

$$t_m = \pm \left\{ \frac{-k_{mm} + [k_{mm}^2 + 16k_{mmm}\epsilon(0)]^{1/2}}{4k_{mmm}} \right\}^{1/2}, \quad (\text{A6})$$

where the sign of  $t_m$  is chosen to be opposite that of  $B_{m,3N}$ .

### B. Morse models

For the Morse models, the potential of Eq. (A1) is replaced by

$$V(u) = D_e(e^{-\beta_e u} - 1)^2. \quad (\text{A7})$$

In Morse approximation I,  $D_e$  is set to the actual dissociation energy for the vibrational potential on the polyatomic surface (cf. Sec. IIIA)

$$D_{eI} = D - V_{MEP}, \quad (\text{A8})$$

while  $\beta_e$  is chosen such that the Morse potential has the correct second derivative at  $u=0$ :

$$\beta_{eI} = (k_{mm}/2D_e)^{1/2}. \quad (\text{A9})$$

In Morse approximation II,  $D_e$  and  $\beta_e$  are determined such that the Morse potential has the correct second and third derivatives at  $u=0$ :

$$D_{eII} = k_{mm}^3/8k_{mmm}, \quad (\text{A10})$$

$$\beta_{eII} = 2|k_{mmm}|/k_{mm}. \quad (\text{A11})$$

We have found that in some cases  $D_{eII} < D_{eI}$ , so that neither the Morse potential nor the energy levels  $\epsilon(n)$  ever reach  $D$ . Thus, in Morse approximation III, we use Morse approximation II if  $D_{eII} > D_{eI}$ ; otherwise, we use Morse approximation I. Given the Morse parameters  $D_e$  and  $\beta_e$ , the energy levels are given by

$$\epsilon(n) = \hbar\omega_e(n + \frac{1}{2})[1 - x_e(n + \frac{1}{2})], \quad (\text{A12})$$

where

$$x_e = \hbar\omega_e/4D_e. \quad (\text{A13})$$

The concave-side turning point of the ground-state level is found as follows: Define

$$z = \pm [\epsilon(0)/D_e]^{1/2}, \quad (\text{A14})$$

where the sign of  $z$  is given by

$$\text{sign}(z) = -\text{sign}(B_{m,3N})\text{sign}(k_{mmm}). \quad (\text{A15})$$

We then have

$$t_m = \frac{1}{\beta_e} \ln(1+z) \text{sign}(k_{mmm}). \quad (\text{A16})$$

The combination of Morse model I or III, when  $k_{mmm} \neq 0$ , with the mixed quadratic-quartic model, when  $k_{mmm} = 0$ , leads to Morse quadratic-quartic models I and III, respectively.

### III. Corrected Dunham $x$ model

Suppose that the potential of Eq. (A1) is truncated at the quartic term, i.e., we take

$$V(u) = \frac{1}{2}k_{mm}u^2 + k_{mmm}u^3 + k_{mmmm}u^4. \quad (\text{A17})$$

Following Dunham,<sup>36</sup> the quantized energies for this potential may be approximated by

$$\epsilon(n) = Y_{00} + Y_{10}(n + \frac{1}{2}) + Y_{20}(n + \frac{1}{2})^2 + Y_{30}(n + \frac{1}{2})^3 + \dots, \quad (\text{A18})$$

where

$$Y_{00} = \frac{a}{8} \left( 3b_2 - \frac{7}{4}b_1^2 \right), \quad (\text{A19})$$

$$Y_{10} = \hbar\omega_e, \quad (\text{A20})$$

$$Y_{20} = \frac{3a}{2} \left( b_2 - \frac{5}{4}b_1^2 \right), \quad (\text{A21})$$

$$Y_{30} = \frac{a^2}{2\hbar\omega_e} \left( -\frac{17}{2}b_2^2 + \frac{225}{4}b_1^2b_2 - \frac{705}{32}b_1^4 \right), \quad (\text{A22})$$

etc., with

$$a = \hbar^2(2\mu)^{-1}, \quad (\text{A23})$$

$$b_1 = 2k_{mmm}/k_{mm}, \quad (\text{A24})$$

and

$$b_2 = 2k_{mmmm}/k_{mm}. \quad (\text{A25})$$

In Eqs. (A19)–(A22), terms on the order of  $a^2/\hbar^2\omega_e^2$ , which corresponds to  $B_e^2/\omega_e^2$  in Dunham's notation,<sup>36</sup> have been dropped. Because the potential (A17) matches that of the true potential energy surface through quartic terms, terms higher than cubic in Eq. (A18) have also been neglected. Even so, there is no simple way to deal with cases where the energy levels reach a maximum in  $n$  before reaching the energy  $D - V_{MEP}$ . We have therefore chosen an alternate energy level expression given by

$$\epsilon(n) = Y_{00} + Y_{10}(n + \frac{1}{2}) + Y'_{20}(n + \frac{1}{2})^2, \quad (\text{A26})$$

where we define

$$Y'_{20} = Y_{20} + \frac{1}{2}Y_{30}, \quad (\text{A27})$$

so that Eqs. (A26) and (A18) give the same result for the zero-point energy through cubic terms. In the event that Eq. (A26) leads to  $\epsilon(n)$  that reaches a maximum below  $D - V_{MEP}$ , we force  $\epsilon(n)$  to maximize at  $D - V_{MEP}$  by setting

$$Y'_{20} = -\frac{\hbar^2\omega_e^2}{4(D - V_{MEP} - Y_{00})}. \quad (\text{A28})$$

The truncation of the energy level series at an  $(n + \frac{1}{2})^2$  term corresponds to truncating the usual spectroscopic expansion  $(\omega_e, x_e, y_e, \dots)^{32}$  at the  $x_e$  term. Because we

correct the coefficient of  $(n + \frac{1}{2})^2$  as in Eq. (A27) or (A28), we call this procedure the corrected Dunham  $x$  method. To obtain the zero-point-level turning point on the concave side of the MEP, we numerically solve the equation

$$V(t_m) = \epsilon(0), \quad (\text{A29})$$

where  $V$  is given by Eq. (A17). The proper root is obtained by initiating the numerical search with the turning point for the corresponding mixed quadratic-quartic potential discussed in Sec. IIA.

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